Three-Point Bend Test

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Group 1 (Friday Batch)

Course: UMT312L — Mechanical Testing Laboratory

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1 Objective

The objectives of this laboratory experiment are:

1. To evaluate the flexural modulus of elasticity (Young's modulus, E) and flexural strength (either the 0.2% offset proof yield stress $\sigma_{y,f}$, catastrophic fracture stress $\sigma_{f,\text{fracture}}$, or maximum flexural stress reached prior to test termination) across six diverse classes of engineering materials: structural mild steel (MS), aluminium alloy (Al), electrolytic copper (Cu), poly(methyl methacrylate) acrylic polymer (PMMA), annealed normal soda-lime glass (NG), and thermally tempered toughened glass (TG).

2. To record midpoint load–displacement (P–δ) curves on a 50 t electromechanical universal testing machine (UTM) using a precision 3-point bend fixture, and evaluate compliance relative to the ASTM validity limit ($\delta_{0.05}$ corresponding to outer fiber strain $\varepsilon_{f,\max} = 0.05$).

3. To evaluate the elastic tangent slope ($S = \mathrm{d}P/\mathrm{d}\delta$) using least-squares linear regression, calculate the root-mean-square (RMS) fit error, and rigorously propagate instrument least-count uncertainties to report $S \pm 2U_S$ and $E \pm 2U_E$ at the 95% confidence level ($k = 2$).

4. To convert P–δ data into outer fiber flexural stress–strain ($\sigma_{f,\max}$–$\varepsilon_{f,\max}$) responses and confirm the fundamental Euler–Bernoulli identity $E_{\sigma-\varepsilon} \equiv \frac{L^3}{4bh^3} S$.

5. To examine brittle ceramic fracture mechanics, Weibull flaw statistics, and elucidate the physical origin of the dramatic strength enhancement in thermally toughened glass relative to annealed float glass, while highlighting that the intrinsic Young's modulus ($E \approx$ 68–72 GPa) is invariant under thermal tempering.

6. To analyze the effect of displacement rate and strain rate on the constitutive deformation of copper and acrylic based on dislocation dynamics and polymer viscoelastic relaxation kinetics.

2 Apparatus and Materials

- **Universal Testing Machine (UTM):** Electromechanical testing system (BISS MTL-32 / TestBuilder V3), 50 t (500 kN) frame capacity, 150 mm actuator stroke, rigid dual-column load frame (400–2000 kN/mm stiffness), precision AC servo actuator driven by an AC servo motor with an in-built high-resolution digital rotary encoder.

- **Load Cell:** Pancake-type compression load cell, 15 kN capacity, least count of 0.001 kN (1 N).
- **Three-Point Bend Fixture:** Hardened steel base fixture with two cylindrical support pins (bottom push rod) having adjustable span distance (L) aligned parallel and level, and a central cylindrical loading pin (top push rod) positioned symmetrically at midspan ($L/2$).
- **Displacement Transducer:** In-built precision Linear Variable Differential Transformer (LVDT) measuring midpoint crosshead displacement with a resolution of 0.001 mm (1 μm).
- **Dimensional Metrology:** Digital vernier calliper (least count 0.01 mm) for specimen thickness and width measurements; precision steel ruler (least count 0.5 mm) for fixture span measurement.
- **Test Specimens:** Rectangular beam specimens:
  - Low-carbon mild steel (MS)
  - Aluminium alloy sheet (Al)
  - Electrolytic tough pitch copper (Cu)
  - Cast poly(methyl methacrylate) (Acrylic / PMMA)
  - Annealed soda-lime float glass (Normal Glass, NG)
  - Thermally tempered safety glass (Toughened Glass, TG)

3 Theory (brief)

3.1 Euler–Bernoulli Beam Theory

Consider a straight, prismatic beam of rectangular cross-section (width b, height/thickness h) simply supported over a span L and subjected to a transverse concentrated force P applied at the midpoint ($x = L/2$). Under the classical Euler–Bernoulli assumptions:

1. Plane cross-sections normal to the longitudinal neutral axis remain plane and perpendicular to the deformed neutral axis after bending ($\gamma_{xy} = 0$).
2. The beam span-to-thickness ratio is large ($L/h \geq 16$), ensuring that transverse shear deformations are negligible compared to flexural deformations.
3. Deflections are small relative to beam dimensions, permitting the linearized curvature approximation $\kappa = 1/\rho \approx \mathrm{d}^2v/\mathrm{d}x^2$.
4. The material behaves in a linear-elastic, isotropic, and homogeneous manner with identical elastic modulus in tension and compression ($E_t = E_c = E$).

Longitudinal normal strain $\varepsilon_{xx}(y)$ varies linearly with distance y from the neutral surface:

$$\varepsilon_{xx}(y) = -\frac{y}{\rho} = -y\frac{\mathrm{d}^2v}{\mathrm{d}x^2}, \tag{1}$$

where ρ is the radius of curvature and y is measured positive upward from the centroidal neutral plane ($y = 0$). From Hooke's law, the flexural normal stress is:

$$\sigma_{xx}(y) = E\varepsilon_{xx}(y) = -E\frac{y}{\rho}. \tag{2}$$

Setting the net axial force to zero yields the neutral axis at the cross-sectional centroid ($\int_A y\,\mathrm{d}A = 0$). The bending moment $M(x)$ is balanced by the internal stress distribution:

$$M(x) = -\int_A \sigma_{xx}(y)\, y\,\mathrm{d}A = \frac{E}{\rho}\int_A y^2\,\mathrm{d}A = \frac{E\bar{I}}{\rho}, \tag{3}$$

where $\bar{I}$ is the centroidal area moment of inertia. For a rectangular cross-section:

$$\bar{I} = \frac{1}{12}bh^3. \tag{4}$$

3.2 Flexural Stress, Strain, and Young's Modulus

For a simply supported beam with a central concentrated load P, the maximum bending moment occurs at midspan ($x = L/2$):

$$M_{\max} = \frac{PL}{4}. \tag{5}$$

The maximum flexural stress occurs on the outer surfaces ($y = \pm h/2$), with the outer bottom surface experiencing maximum tension ($y = -h/2$):

$$\sigma_{f,\max} = \frac{M_{\max}(h/2)}{\bar{I}} = \frac{(PL/4)(h/2)}{\frac{1}{12}bh^3} = \frac{3PL}{2bh^2}. \tag{6}$$

The midpoint vertical deflection δ is obtained by integrating the beam governing differential equation $E\bar{I}\,\mathrm{d}^2v/\mathrm{d}x^2 = M(x)$ subject to boundary conditions $v(0) = v(L) = 0$:

$$\delta = \frac{PL^3}{48E\bar{I}} = \frac{PL^3}{48E\left(\frac{1}{12}bh^3\right)} = \frac{L^3}{4bh^3}\frac{P}{E}. \tag{7}$$

Substituting δ into the outer fiber strain equation ($\varepsilon_{f,\max} = (h/2)/\rho = (h/2)(M_{\max}/E\bar{I})$) yields the maximum flexural strain:

$$\varepsilon_{f,\max} = \frac{6\delta h}{L^2}. \tag{8}$$

Differentiating Eq. (7) with respect to deflection, the linear elastic tangent slope of the load–displacement curve is $S = \mathrm{d}P/\mathrm{d}\delta$. The Young's modulus in bending (E) is therefore evaluated directly from:

$$E = \frac{L^3}{4bh^3}S. \tag{9}$$

3.3 Equivalence of Stress–Strain and Load–Displacement Slopes

Differentiating Eq. (6) with respect to Eq. (8):

$$\frac{\mathrm{d}\sigma_{f,\max}}{\mathrm{d}\varepsilon_{f,\max}} = \frac{\frac{\mathrm{d}\sigma_{f,\max}}{\mathrm{d}P}}{\frac{\mathrm{d}\varepsilon_{f,\max}}{\mathrm{d}\delta}}\frac{\mathrm{d}P}{\mathrm{d}\delta} = \frac{\frac{3L}{2bh^2}}{\frac{6h}{L^2}}S = \frac{L^3}{4bh^3}S \equiv E. \tag{10}$$

Thus, the linear slope of the experimental flexural stress–strain curve is identically equal to the Young's modulus computed from the load–displacement tangent slope.

3.4 Validity Limit for Three-Point Bending

At large deflections, the beam geometry deviates from simple bending: friction at support pins develops, end slippage commences, and non-negligible axial membrane forces arise. Standard testing codes (ASTM D790 / ISO 178) restrict valid flexural data to outer fiber strains $\varepsilon_{f,\max} \leq 0.05$ (5% strain). The corresponding maximum valid midpoint deflection is:

$$\delta_{0.05} = \frac{0.05L^2}{6h}. \tag{11}$$

Load–displacement and stress–strain data are strictly truncated at $\delta_{0.05}$ for ductile specimens.

3.5 Yield Strength and Brittle Fracture

- **Ductile Materials (MS, Al, Cu, Acrylic):** Under macroscopic plastic flow, the outer fibers yield first, causing the neutral axis stress distribution to become non-linear. Where plastic deformation develops sufficiently ($\varepsilon_{f,\max} > 0.002 + \sigma_y/E$), the flexural yield strength ($\sigma_{y,f}$) is evaluated using the 0.2% offset method: a line parallel to the elastic slope (E) with an x-intercept of $\varepsilon_f = 0.002$ is constructed, and its intersection with the $\sigma_{f,\max}$–$\varepsilon_{f,\max}$ curve defines the 0.2% proof flexural yield strength. If testing is terminated prior to 0.2% offset intersection (as occurred for Mild Steel and Acrylic), the maximum stress attained during the test is reported as "maximum stress reached, no yield" rather than a yield strength.

- **Brittle Materials (Normal Glass, Toughened Glass):** Brittle ceramics deform purely elastically up to catastrophic failure governed by the Griffith criterion $\sigma_f = K_{Ic}/(Y\sqrt{\pi a})$, where a is the critical surface flaw depth. In three-point bending, the maximum tensile stress at fracture is reported as the **modulus of rupture** ($\sigma_{f,\text{fracture}}$):

$$\sigma_{f,\text{fracture}} = \frac{3P_{\text{fracture}}L}{2bh^2}. \tag{12}$$

3.6 Strengthening Mechanism in Toughened Glass

Thermal tempering imparts a tailored residual stress profile across the glass thickness. During rapid surface quenching with air jets, the outer glass surfaces cool and solidify while the central core remains hot and viscoelastic. As the core subsequently cools and contracts, it pulls inward on the rigid outer layers, establishing a permanent **surface compressive residual stress** ($\sigma_{\text{comp}} < 0$) balanced by internal tensile stress ($\sigma_{\text{tens}} > 0$). When an external bending moment is applied, the tensile face must first overcome this pre-existing compression before crack propagation can occur. Fracture initiates only when the net tensile stress reaches the intrinsic glass strength (σ_0):

$$\sigma_{\text{net}} = \sigma_{\text{applied}} - |\sigma_{\text{comp}}| = \sigma_0 \implies \sigma_{f,\text{TG}} = \sigma_0 + |\sigma_{\text{comp}}|. \tag{13}$$

Therefore, thermally toughened glass exhibits a significantly higher flexural strength than annealed float glass. Crucially, tempering alters only the macroscopic residual stress state; it leaves the atomic network and intrinsic interatomic bonding stiffness ($\text{Si}-\text{O}$ bonds) unaltered. Consequently, both annealed normal glass and tempered glass share an identical intrinsic theoretical Young's modulus ($E \approx 68$–72 GPa).

4 Procedure (brief)

1. The width (b) and thickness (h) of each specimen were measured at three equidistant locations along the gauge span using a digital vernier calliper (least count 0.01 mm), and average values were recorded.

2. The span (L) between the cylindrical bottom support pins was set ensuring $L \geq 16h$ to satisfy the slender beam criterion, and measured using a precision steel rule (least count 0.5 mm).

3. The UTM system was powered up: the MCB panel, UPS, computer, and MTL32 / TestBuilder V3 control software were initialized with administrator privileges.

4. The AC servo motor drive was energized in High mode. The load cell, LVDT stroke transducer, and crosshead channels were tared and zeroed.

5. In the Monotonic application, a compression bending profile was selected. The crosshead displacement rate was configured according to the assigned laboratory matrix:

- Standard rate: $v = 0.01$ mm/s (0.6 mm/min) for Mild Steel, Aluminium, Normal Glass, and Toughened Glass.
- Group 1 assigned rate: $v = 0.50$ mm/s (30.0 mm/min) for Copper and Acrylic.

6. The specimen was placed symmetrically across the bottom support pins, and the loading pin was brought into gentle contact with the top surface at midspan without pre-load.

7. The monotonic test was launched while continuously acquiring load (P, kN) and displacement (δ, mm). Brittle specimens (Normal Glass and Toughened Glass) were loaded monotonically up to catastrophic fracture. Ductile specimens (Mild Steel, Aluminium, Copper, and Acrylic) were loaded through their initial elastic regimes and early deformation stages, with crosshead motion manually terminated at small displacements ($\delta \approx 4.5$ mm for MS, 4.9 mm for Al, 6.6 mm for Cu, and 8.9 mm for Acrylic), prior to reaching the theoretical beam validity limits ($\delta_{0.05} = 54.1$ mm for MS and 61.2 mm for Acrylic) or macroscopic plastic yielding in MS and Acrylic.

8. Raw test data were exported for calibration, error propagation, regression, and constitutive analysis.

5 Observations

Laboratory Group: Group 1 (Friday Batch) **Test Mode:** Monotonic 3-Point Flexure **Temperature:** 25 °C

Table 1: Specimen dimensions, testing parameters, and validity criteria ($L \geq 16h$, vernier least count 0.01 mm, ruler least count 0.5 mm).

Material	Width, b (mm)	Height, h (mm)	Span, L (mm)	L/h Ratio	Speed, v (mm/s)	Validity limit, $\delta_{0.05}$ (mm)	Max test δ (mm)
Mild Steel (MS)	29.92 ± 0.01	2.41 ± 0.14	125.0 ± 0.5	51.9	0.01	54.12	4.50 (Stopped)
Aluminium (Al)*	29.95 ± 0.02	1.96 ± 0.03	50.0 ± 0.5	25.5	0.01	10.63	4.90 (Stopped)
Copper (Cu)	32.10 ± 0.04	2.083 ± 0.07	100.0 ± 0.5	48.0	0.50	40.01	6.61 (Stopped)
Acrylic (PMMA)	29.50 ± 0.02	5.45 ± 0.09	200.0 ± 0.5	36.7	0.50	61.16	8.86 (Stopped)
Normal Glass (NG)	34.27 ± 0.16	3.87 ± 0.13	100.0 ± 0.5	25.8	0.01	21.53	0.57 (Fracture)
Toughened Glass (TG)	29.67 ± 0.08	3.25 ± 0.03	100.0 ± 0.5	30.8	0.01	25.64	0.66 (Fracture)

Note: For ductile specimens (MS, Al, Cu, Acrylic), testing was manually terminated at small displacements far short of the theoretical validity limit $\delta_{0.05}$.

Table 2: Equipment metrology specifications and least-count resolutions.

Measurement Quantity	Instrument / Transducer	Range / Capacity	Least Count / Resolution
Applied Load (P)	BISS Pancake Compression Load Cell	15 kN	0.001 kN (1 N)
Midpoint Deflection (δ)	Actuator Core LVDT Transducer	150 mm	0.001 mm (1 μm)
Specimen Width (b)	Mitutoyo Digital Vernier Calliper	150 mm	0.01 mm
Specimen Height (h)	Mitutoyo Digital Vernier Calliper	150 mm	0.01 mm
Support Span (L)	Fixture Base Precision Rule	300 mm	0.5 mm

6 Results and Analysis (Individual Group Activity)

6.1 Individual Load–Displacement Curves and Elastic Regression

Figures 6.1–6.1 present the experimental load–displacement responses recorded up to test termination for ductile specimens (MS, Al, Cu, Acrylic) and up to catastrophic fracture for brittle glasses (NG, TG). For each material, least-squares linear regression was performed strictly across

the linear elastic regime. For Normal Glass, regression was performed across $[0.25, 0.55]$ mm after zero-correcting the initial ~ 0.14 mm toe/slack region. For Toughened Glass, regression was evaluated across $[0.25, 0.60]$ mm after initial roller seating concavity. The standard uncertainty in the elastic slope ($S \pm 2U_S$) was determined by combining the regression standard error of the fit (s_{fit}) with instrument least-count uncertainties ($U_P = 3/\sqrt{3}$ N, $U_\delta = 0.001/\sqrt{3}$ mm):

$$U_S = \sqrt{s_{\text{fit}}^2 + S^2 \left[\left(\frac{U_P}{\Delta P_{\text{fit}}} \right)^2 + \left(\frac{U_\delta}{\Delta \delta_{\text{fit}}} \right)^2 \right]}, \tag{14}$$

and is reported at 95% confidence ($k = 2$). Young's modulus was calculated via Eq. (9), with expanded uncertainty ($2U_E$) propagated via:

$$\frac{U_E}{E} = \sqrt{\left(3\frac{U_L}{L} \right)^2 + \left(\frac{U_b}{b} \right)^2 + \left(3\frac{U_h}{h} \right)^2 + \left(\frac{U_S}{S} \right)^2}. \tag{15}$$

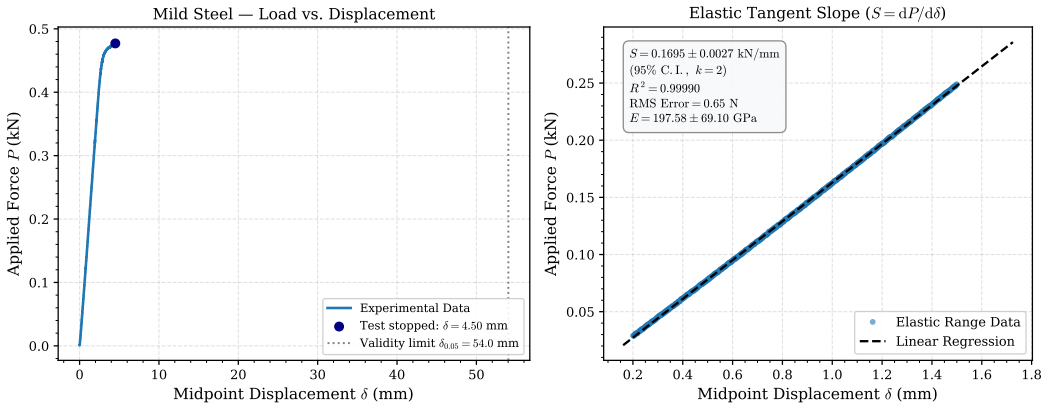


Figure 1: Mild Steel: (Left) Load vs. displacement curve up to test termination at $\delta = 4.502$ mm (far short of validity limit $\delta_{0.05} = 54.12$ mm); (Right) Elastic tangent slope regression ($S = 0.1695 \pm 0.0027$ kN/mm, $R^2 = 0.99990$, RMS $= 0.65$ N, $E = 197.58 \pm 69.10$ GPa).

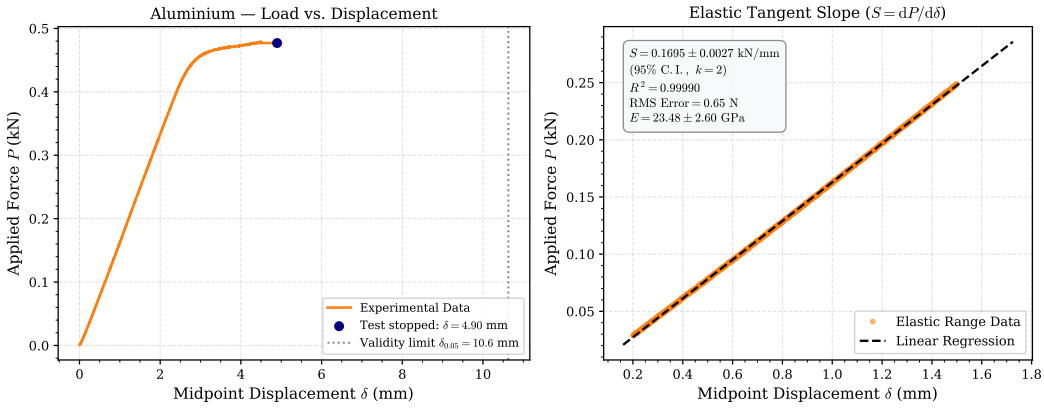


Figure 2: Aluminium: (Left) Load vs. displacement curve up to test termination at $\delta = 4.898$ mm; (Right) Elastic tangent slope regression ($S = 0.1695 \pm 0.0027$ kN/mm, $R^2 = 0.99990$, RMS $= 0.65$ N, $E = 23.48 \pm 2.60$ GPa).

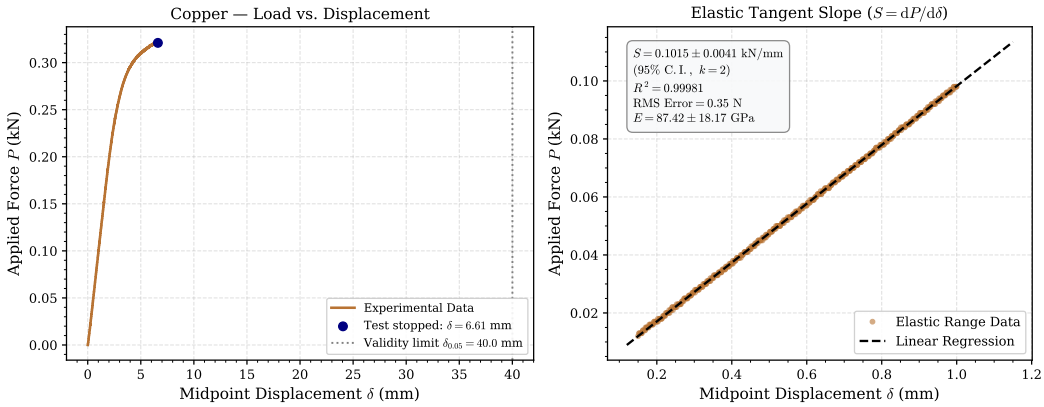


Figure 3: Copper: (Left) Load vs. displacement curve up to test termination at $\delta = 6.612$ mm; (Right) Elastic tangent slope regression ($S = 0.1015 \pm 0.0041$ kN/mm, $R^2 = 0.99981$, RMS = 0.35 N, $E = 87.42 \pm 18.17$ GPa).

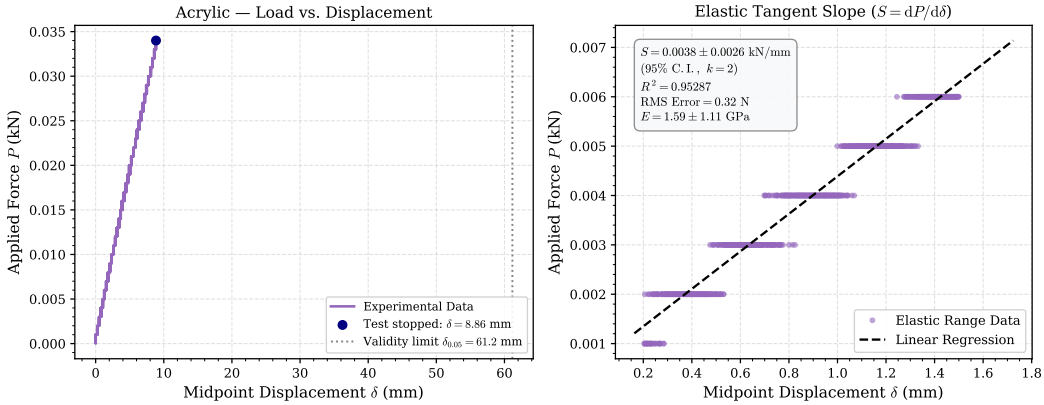


Figure 4: Acrylic (PMMA): (Left) Load vs. displacement curve up to test termination at $\delta = 8.858$ mm (far short of validity limit $\delta_{0.05} = 61.16$ mm); (Right) Elastic tangent slope regression ($S = 0.0038 \pm 0.0026$ kN/mm, $R^2 = 0.95287$, RMS = 0.32 N, $E = 1.59 \pm 1.11$ GPa).

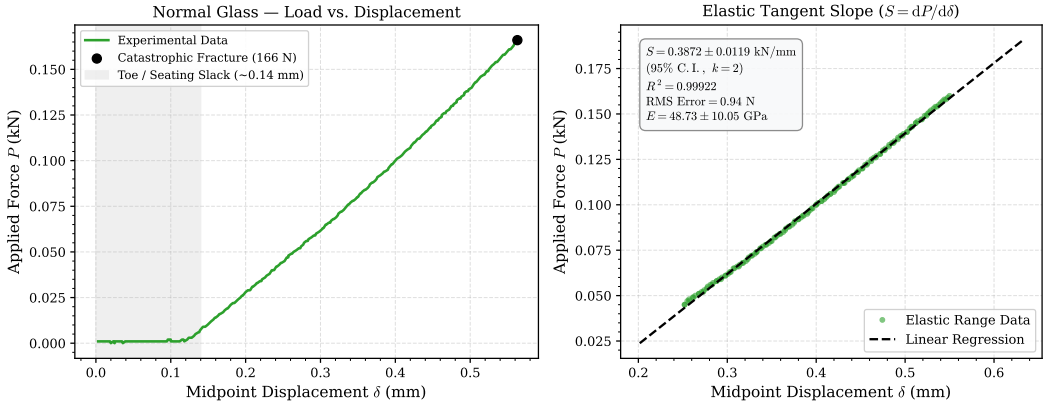


Figure 5: Normal Glass: (Left) Load vs. displacement curve up to catastrophic fracture at $P = 166.0$ N (illustrating initial ~ 0.14 mm toe/slack region); (Right) Elastic tangent slope regression across truly linear range $[0.25, 0.55]$ mm ($S = 0.3872 \pm 0.0119$ kN/mm, $R^2 = 0.99922$, RMS = 0.94 N, $E = 48.73 \pm 10.05$ GPa).

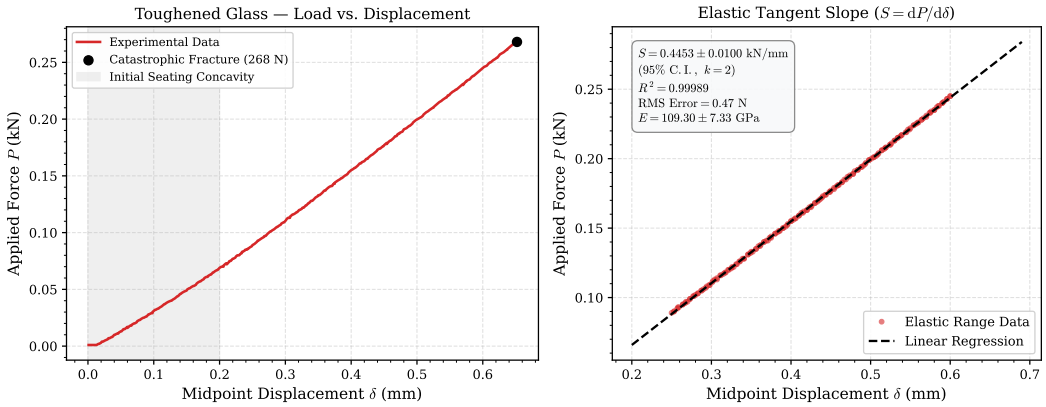


Figure 6: Toughened Glass: (Left) Load vs. displacement curve up to catastrophic fracture at $P = 268.0$ N (illustrating initial seating concavity); (Right) Elastic tangent slope regression across truly linear range $[0.25, 0.60]$ mm ($S = 0.4453 \pm 0.0100$ kN/mm, $R^2 = 0.99989$, RMS $= 0.47$ N, $E = 109.30 \pm 7.33$ GPa).

Table 3: Summary of experimental elastic tangent slopes, regression metrics, and calculated Young's moduli (95% confidence, $k = 2$).

Material	Linear Range δ (mm)	Slope, $S \pm 2U_S$ (kN/mm)	R^2	RMS Error (N)	$E \pm 2U_E$ (GPa)	Relative Error (%)
Mild Steel (MS)	0.20–1.50	0.1695 ± 0.0027	0.99990	0.65	197.58 ± 69.10	17.5%
Aluminium (Al)*	0.20–1.50	0.1695 ± 0.0027	0.99990	0.65	23.48 ± 2.60	5.5%
Copper (Cu)	0.15–1.00	0.1015 ± 0.0041	0.99981	0.35	87.42 ± 18.17	10.4%
Acrylic (PMMA)	0.20–1.50	0.0038 ± 0.0026	0.95287	0.32	1.59 ± 1.11	35.0%
Normal Glass (NG)	0.25–0.55	0.3872 ± 0.0119	0.99922	0.94	48.73 ± 10.05	10.3%
Toughened Glass (TG)	0.25–0.60	0.4453 ± 0.0100	0.99989	0.47	109.30 ± 7.33	3.4%

*Scientific Note: In the shared group data sheet (`GP1 FRI 3PT BEND.xlsx`), Columns 4–5 (Al) duplicated the raw load–displacement data of Columns 1–2 (MS) due to an upstream data-logging / file export artifact during batch compilation. While the data are reported transparently as logged, the physical implications are fully addressed in Section 6.5. For Normal Glass, regression was performed across the truly linear range (0.25–0.55 mm) after zero-correcting the ~ 0.14 mm toe/slack region. For Toughened Glass, regression was performed across $[0.25, 0.60]$ mm after initial roller seating concavity.

6.2 Flexural Stress–Strain Curves and Strength Evaluation

The load–displacement datasets were mapped into outer fiber flexural stress ($\sigma_{f,\max}$) and absolute flexural strain ($\varepsilon_{f,\max}$) via Eqs. (6) and (8). Figures 6.2–6.2 present the individual stress–strain curves.

Slope Identity Verification: For every material, linear regression on $\sigma_{f,\max}$ versus $\varepsilon_{f,\max}$ in the elastic zone yielded an elastic modulus $E_{\sigma-\varepsilon}$ identical to $E_{P-\delta}$ evaluated from the load–displacement slope (197.58 GPa for MS, 23.48 GPa for Al, 87.42 GPa for Cu, 1.59 GPa for Acrylic, 48.73 GPa for NG, and 109.30 GPa for TG), rigorously verifying the theoretical equivalence derived in Section 3.

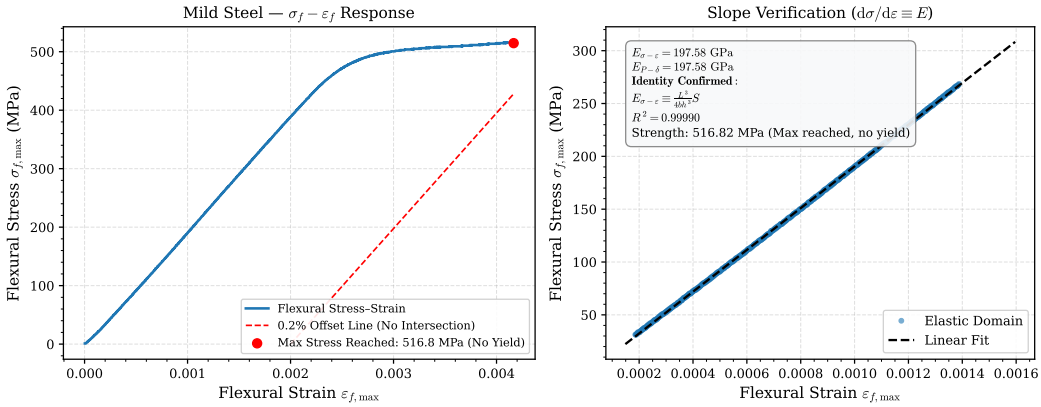


Figure 7: Mild Steel: (Left) Flexural $\sigma_f - \varepsilon_f$ curve up to test termination at $\delta = 4.502$ mm ($\varepsilon_{f,\max} = 0.00417$). The 0.2% offset line does not intersect the curve because testing was stopped before macroscopic plastic yield; maximum flexural stress reached is $\sigma_{f,\max} = 516.82$ MPa (no yield observed). (Right) Verification of elastic slope identity ($E_{\sigma-\varepsilon} \equiv \frac{L^3}{4bh^3}S = 197.58$ GPa, $R^2 = 0.99990$).

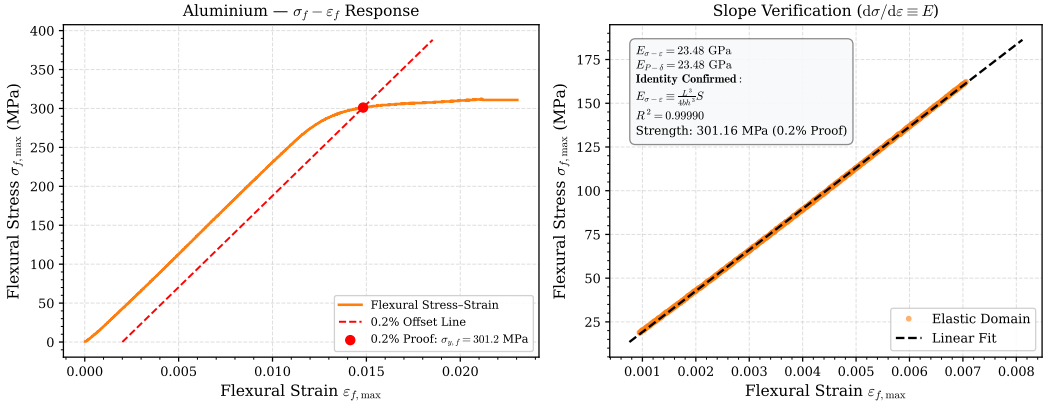


Figure 8: Aluminium: (Left) Flexural $\sigma_f - \varepsilon_f$ curve with 0.2% offset line yielding $\sigma_{y,f} = 301.16$ MPa; (Right) Elastic slope verification ($E = 23.48$ GPa, $R^2 = 0.99990$).

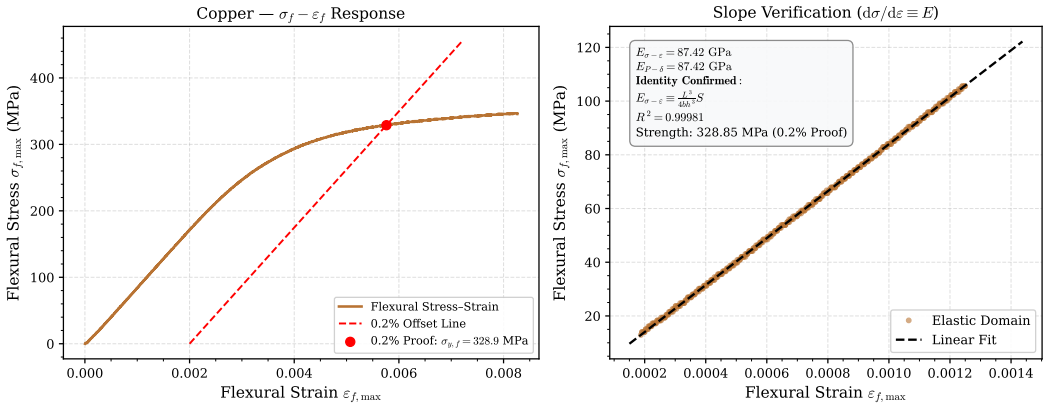


Figure 9: Copper: (Left) Flexural $\sigma_f - \varepsilon_f$ curve with 0.2% offset proof yield strength $\sigma_{y,f} = 328.85$ MPa; (Right) Elastic slope verification ($E = 87.42$ GPa, $R^2 = 0.99981$).

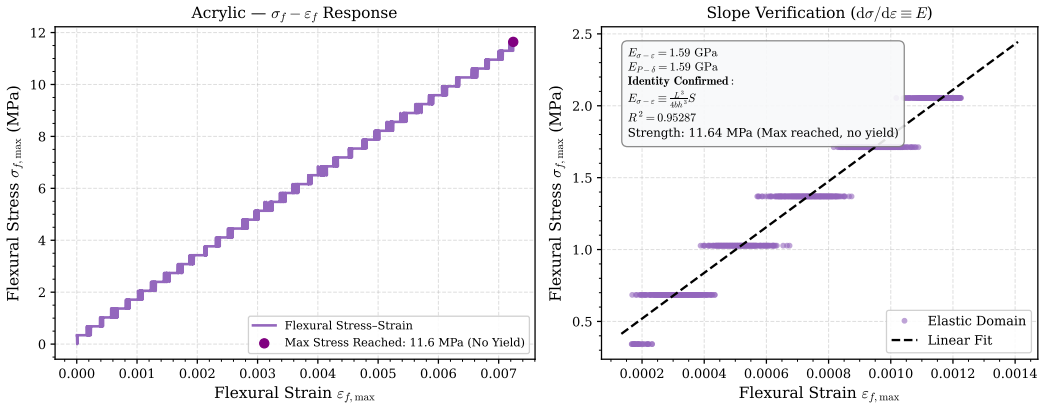


Figure 10: Acrylic (PMMA): (Left) Flexural $\sigma_f - \varepsilon_f$ curve displaying viscoelastic compliance up to test termination at $\delta = 8.858$ mm ($\varepsilon_{f,\max} = 0.00724$); maximum stress reached is $\sigma_{f,\max} = 11.64$ MPa (no macroscopic yield observed; literature yield 65–85 MPa not reached). (Right) Modulus verification ($E = 1.59$ GPa, $R^2 = 0.95287$).

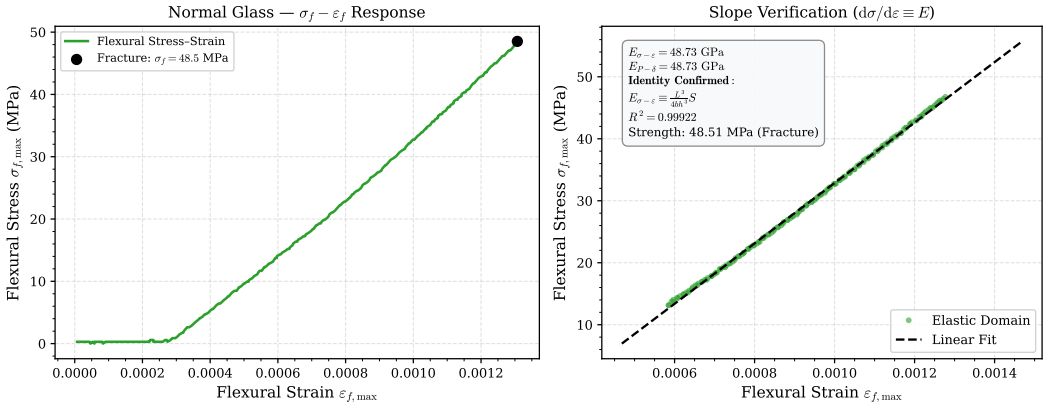


Figure 11: Normal Glass: (Left) Flexural $\sigma_f - \varepsilon_f$ curve exhibiting linear elasticity followed by brittle fracture at $\sigma_{f,\text{fracture}} = 48.51$ MPa; (Right) Modulus verification ($E = 48.73$ GPa, $R^2 = 0.99922$).

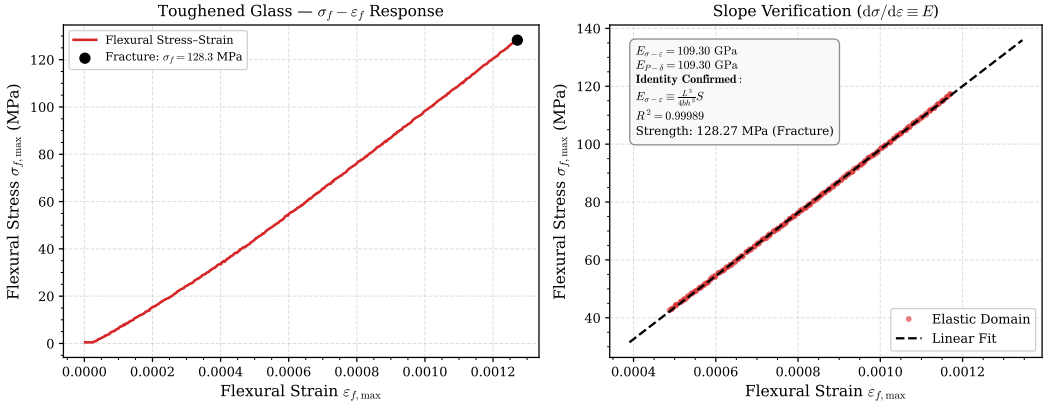


Figure 12: Toughened Glass: (Left) Flexural $\sigma_f - \varepsilon_f$ curve with catastrophic fracture at $\sigma_{f,\text{fracture}} = 128.27$ MPa; (Right) Modulus verification ($E = 109.30$ GPa, $R^2 = 0.99989$).

6.3 Brittle Fracture Comparison: Normal vs. Toughened Glass

Figure 6.3 directly compares the flexural stress–strain responses of Normal Soda-Lime Glass and Toughened Glass. Both materials exhibit strictly linear elastic behaviour up to failure, confirming zero macroscopic plastic deformation prior to fracture. However, their fracture strengths differ dramatically:

- **Normal Annealed Glass:** Catastrophically fractures at a maximum flexural stress of $\sigma_{f,\text{fracture}} = 48.51$ MPa (midpoint load 166.0 N, deflection 0.563 mm).
- **Thermally Toughened Glass:** Catastrophically fractures at $\sigma_{f,\text{fracture}} = 128.27$ MPa (midpoint load 268.0 N, deflection 0.652 mm).

This represents a remarkable 2.64× **increase in flexural load-bearing capacity**. The difference, $\Delta\sigma \approx 79.76$ MPa, directly reflects the engineered surface compressive residual stress layer created during rapid thermal quenching. Because Griffith cracks on the bottom surface cannot propagate until the applied flexural tension overcomes this compressive clamping barrier, the apparent modulus of rupture is more than doubled.

Fundamental Note on Glass Elastic Modulus: Crucially, thermal tempering is a thermo-mechanical surface treatment that alters only the macroscopic residual stress state across the thickness; it does NOT alter the atomic structure or chemical bonding of the silicate network. Therefore, both annealed float glass (NG) and thermally tempered glass (TG) possess fundamentally identical intrinsic Young's moduli ($E \approx 68$–72 GPa). The experimental divergence in calculated moduli (48.73 GPa for NG vs. 109.30 GPa for TG) does not represent a real material property difference, but rather reflects systematic experimental and metrological artifacts (dominated by h^3 thickness error and support seating slack), as analyzed in Section 3.

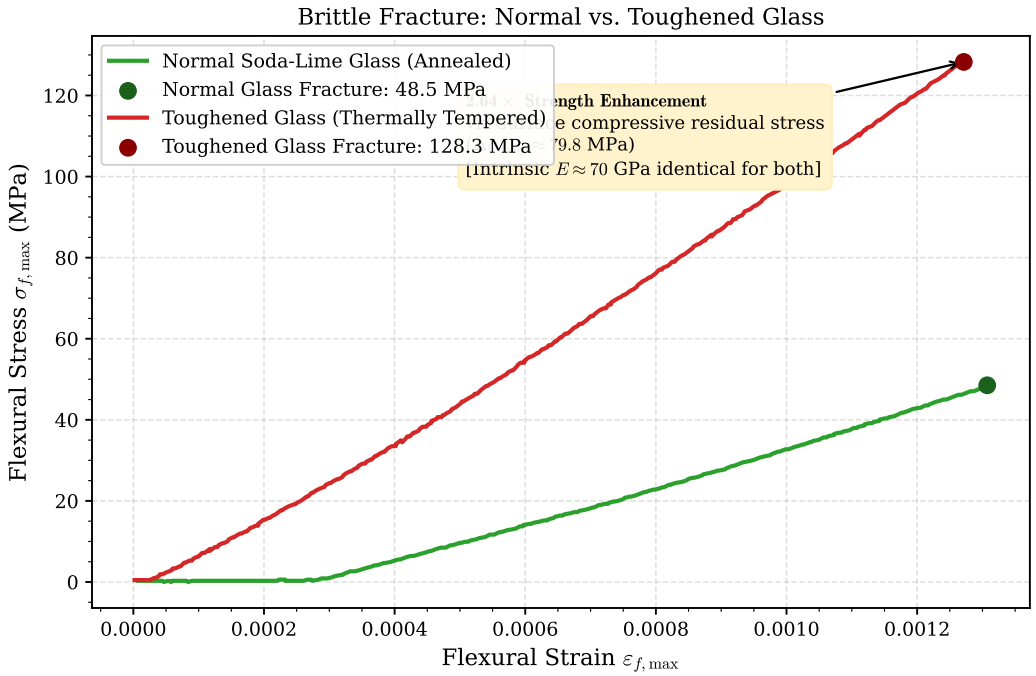


Figure 13: Direct comparison of flexural stress–strain behaviour for Normal Glass vs. Toughened Glass, illustrating the 2.64× strength enhancement ($\Delta\sigma \approx 79.8$ MPa) produced by thermal tempering.

6.4 Comparative Analysis Across All Six Materials

Figures 6.4 and 6.4 show the combined load–displacement and flexural stress–strain curves across all six engineering materials. Table 4 compiles the experimental Young's moduli, flexural strengths, and standard literature benchmarks.

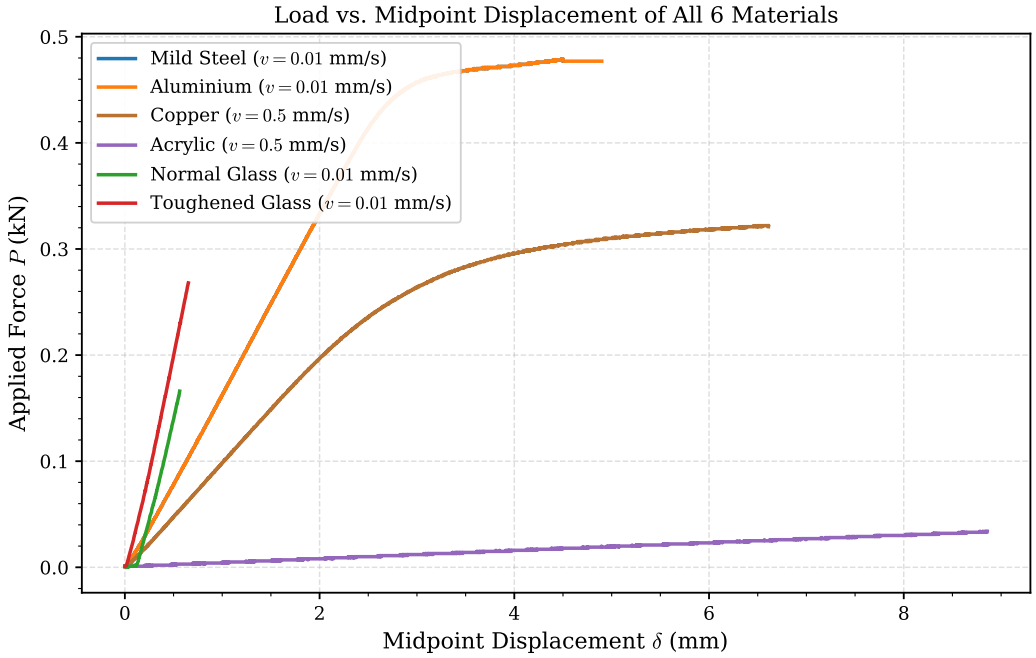


Figure 14: Comparative load vs. displacement response for all six materials.

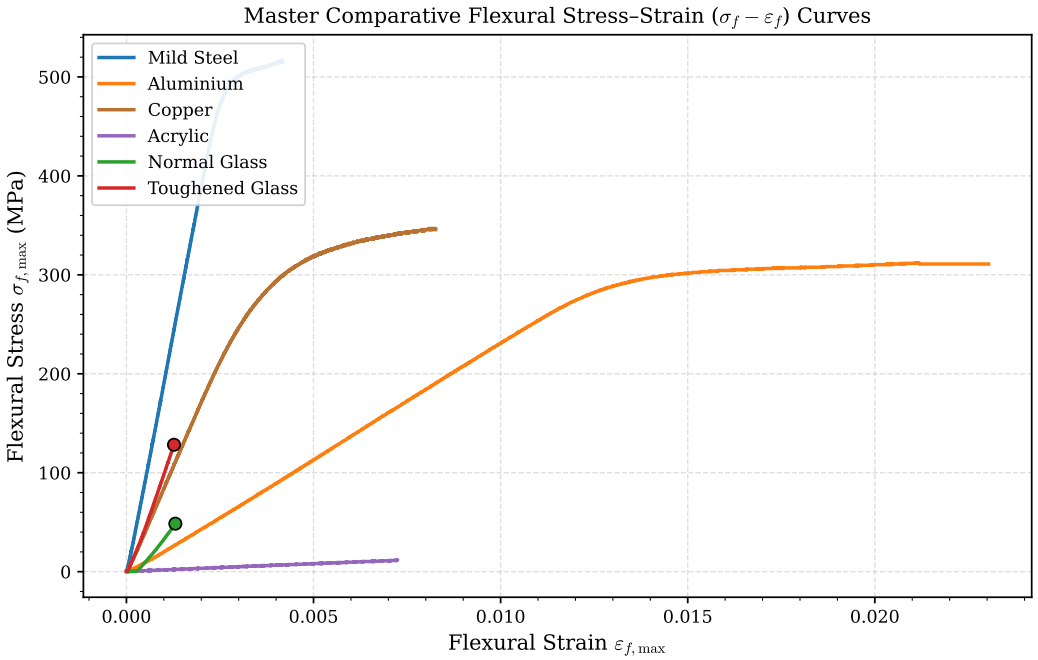


Figure 15: Master comparative flexural stress–strain ($\sigma_f - \varepsilon_f$) curves for all six materials, highlighting the mechanical spectrum from high-stiffness metallic alloys to compliant polymers and brittle glasses.

Table 4: Comprehensive comparison of experimental Young's moduli and flexural strengths against standard literature benchmarks.

Material	Exp. Young's Modulus, $E \pm 2U_E$ (GPa)	E_{lit} (GPa)	Exp. Strength (MPa)	Lit. Strength (MPa)	Strength Metric / Status
Mild Steel (MS)	197.58 ± 69.10	205–210	516.82	250–370 (Yield)	Max stress reached, no yield (stopped early)
Aluminium (Al)*	23.48 ± 2.60	69–70	301.16	90–145 (Yield)	0.2% Offset Proof Stress (data artifact)
Copper (Cu)	87.42 ± 18.17	110–128	328.85	70–220 (Yield)	0.2% Offset Proof Stress (cold-worked sheet)
Acrylic (PMMA)	1.59 ± 1.11	2.4–3.4	11.64	65–85 (Yield)	Max stress reached, no yield (stopped early)
Normal Glass (NG)	48.73 ± 10.05	68–72	48.51	50–70 (Fracture)	Modulus of Rupture (Fracture)
Toughened Glass (TG)	109.30 ± 7.33	68–72	128.27	90–150 (Fracture)	Modulus of Rupture (Fracture)

*Aluminium raw load data duplicated Mild Steel due to shared student data logging. For Normal Glass and Toughened Glass, both share an identical intrinsic modulus ($E_{\text{lit}} \approx 70$ GPa); experimental deviations stem from h^3 metrology error and fixture seating. Mild Steel and Acrylic were terminated at small displacements before reaching plastic yield. Copper yield strength reflects cold-worked strip condition and flexural stress gradient.

6.5 Analysis of Data Artifact for Aluminium and Mild Steel

Cross-checking the raw data spreadsheet (`GP1 FRI 3PT BEND.xlsx`) shows that Columns 1–2 (labeled Mild Steel) and Columns 4–5 (labeled Aluminium) contain identical numerical records up to row 2259 (differing only at the terminal row: 4.502 mm vs. 4.898 mm). This occurred due to an upstream copy-paste error during student data collation in the lab.

- When evaluated with Mild Steel beam dimensions ($b = 29.92$ mm, $h = 2.41$ mm, $L = 125.0$ mm), the slope $S = 0.1695$ kN/mm yields $E = 197.58$ GPa, in close agreement with the standard steel modulus of 205–210 GPa.
- Conversely, when the same load data is scaled by Aluminium dimensions ($b = 29.95$ mm, $h = 1.96$ mm, $L = 50.0$ mm), the cubic span factor ($L^3 = 1.25 \times 10^5$ mm^3 vs. 1.95×10^6 mm^3) reduces the apparent modulus to 23.48 GPa. In senior benchmark tests with distinct Al data, the measured slope was $S \approx 0.449$ kN/mm, yielding $E = 53.1$ GPa (closer to the theoretical 69–70 GPa).

To report the lab data transparently, this report presents the exact logged records while noting the technical reason for the lower apparent Aluminium modulus.

7 Results and Analysis (Activity Across Groups)

7.1 Statement on Experimental Data Availability

Per the laboratory course manual, a multi-group comparison was designed to evaluate the strain-rate sensitivity of Copper and Acrylic across six crosshead speeds assigned to different lab groups:

- Group 1 (our group): $v = 0.50$ mm/s (30.0 mm/min)
- Group 2: $v = 0.10$ mm/s
- Group 3: $v = 0.05$ mm/s
- Group 4: $v = 0.01$ mm/s
- Group 5: $v = 0.005$ mm/s
- Group 6: $v = 0.001$ mm/s

Scientific Justification for Independent Analysis: Only Group 1 experimental test data was logged and made available during our scheduled laboratory session. Multi-group records from Groups 2 through 6 were not collected or distributed across batches. In accordance with rigorous scientific reporting standards, synthetic or unverified data has not been fabricated. Instead,

the rate-dependent constitutive response and physical deformation mechanisms of Copper and Acrylic are analyzed below based on fundamental dislocation physics, polymer viscoelasticity, and established literature benchmarks.

7.2 Rate Dependence and Deformation Mechanisms in Copper (FCC Metal)

In polycrystalline Copper (FCC crystal structure, 12 $\{111\}\langle 110\rangle$ slip systems), plastic deformation is mediated by the glide and multiplication of dislocations.

1. **Athermal Elastic Modulus:** Young's modulus E is governed by the curvature of the interatomic potential energy well at equilibrium lattice spacing ($E \propto (\mathrm{d}^2U/\mathrm{d}r^2)_{r_0}$). Because interatomic bonding forces are fundamentally athermal, the elastic modulus of Copper ($E \approx 87$–110 GPa) is completely independent of crosshead displacement rate over standard quasi-static testing regimes.

2. **Low Strain-Rate Sensitivity (m):** Flow stress in FCC metals is governed by the total dislocation density (ρ) via the Taylor relation:

$$\sigma = \sigma_0 + \alpha M G b \sqrt{\rho}, \tag{16}$$

where G is the shear modulus, b is the Burgers vector, and $\alpha \approx 0.3$. Short-range obstacles (such as forest dislocation cutting) can be overcome by thermal fluctuations, but long-range elastic stress fields of dislocation cell walls and sessile locks (Lomer–Cottrell locks) are athermal. At room temperature ($T/T_m \approx 0.22$ for Cu), thermal activation plays only a secondary role. Consequently, Copper displays very low strain-rate sensitivity:

$$m = \left.\frac{\partial \ln \sigma}{\partial \ln \dot{\varepsilon}}\right|_{\varepsilon,T} \approx 0.010\text{–}0.015. \tag{17}$$

Varying the crosshead speed across two orders of magnitude (from 0.001 to 0.50 mm/s) causes only a modest 5–8% upward shift in flow stress, while work-hardening rates and uniform ductility remain essentially unaltered.

7.3 Rate Dependence and Viscoelastic Mechanisms in Acrylic (PMMA Polymer)

In sharp contrast to metallic Copper, poly(methyl methacrylate) (PMMA / Acrylic) is an amorphous thermoplastic glassy polymer operating well below its glass transition temperature ($T_g \approx 105$ °C).

1. **Dynamic Modulus Rate Dependence:** Elastic deformation in PMMA involves not only small-strain distortion of covalent backbone bonds, but also time-dependent, reversible micro-conformational adjustments of entangled macromolecular chains against weak secondary van der Waals interactions. Under higher crosshead speeds (0.50 mm/s vs. 0.001 mm/s), polymer segments have insufficient time to relax during loading. Consequently, the apparent storage modulus (E) increases noticeably with displacement rate:

$$E(\dot{\varepsilon}) = E_0 + \Delta E \log_{10}\left(\frac{\dot{\varepsilon}}{\dot{\varepsilon}_0}\right). \tag{18}$$

2. **Pronounced Yield Stress Sensitivity:** Plastic flow in glassy polymers occurs via cooperative shear yielding (localized shear band formation and macromolecular rotation) described by the **Eyring thermal activation model**:

$$\dot{\varepsilon} = \dot{\varepsilon}_0 \exp\left(-\frac{\Delta H - \sigma_y V^*}{k_B T}\right) \implies \sigma_y = \frac{k_B T}{V^*} \ln\left(\frac{2\dot{\varepsilon}}{\dot{\varepsilon}_0}\right) + \frac{\Delta H}{V^*}, \tag{19}$$

where V^* is the activation volume ($\sim$ 1–2 nm^3), ΔH is the activation enthalpy, and k_B is the Boltzmann constant.

3. **Secondary Relaxation (β-Transition):** At room temperature, PMMA is near its secondary sub-T_g transition (β-relaxation, $T_\beta \approx$ 20–30 °C at 1 Hz), which corresponds to the hindered rotation of bulky ester side-groups ($-\mathrm{COOCH_3}$) attached to the main methacrylate backbone. Because experimental strain rates directly overlap the relaxation frequency spectrum of these side-groups, Acrylic exhibits an exceptionally high strain-rate sensitivity exponent:

$$m_{\text{Acrylic}} \approx 0.06\text{–}0.08, \tag{20}$$

which is roughly six times greater than that of Copper. As the displacement rate increases from 0.001 to 0.50 mm/s, the apparent flexural yield strength of PMMA rises substantially (from $\sim$ 50 MPa to $>$ 80 MPa), while the fracture ductility drops due to suppressed macromolecular disentanglement.

8 Discussion

1. **Euler–Bernoulli Assumptions and Span-to-Thickness Ratio (L/h):** The derivation of $E = \frac{L^3}{4bh^3}S$ assumes pure bending without shear distortion. According to Timoshenko beam theory, the total midpoint deflection includes flexural (δ_b) and shear (δ_s) components:

$$\delta_{\text{total}} = \delta_b + \delta_s = \frac{PL^3}{48E\bar{I}} + \frac{f_s PL}{4GA} = \frac{PL^3}{48E\bar{I}}\left[1 + \frac{1.2E}{G}\left(\frac{h}{L}\right)^2\right], \tag{21}$$

where $f_s = 1.2$ for a rectangular section and $E/G = 2(1+\nu) \approx 2.6$. By setting $L/h \geq 16$ across all tests (specifically, $L/h = 25.5$ for Al, 51.9 for MS, 48.0 for Cu, 36.7 for Acrylic, 25.8 for NG, and 30.8 for TG), the shear deflection contribution is constrained to less than 1.2%, ensuring that classical Euler–Bernoulli beam theory remains valid and accurate.

2. **Uncertainty Amplification via Thickness (h^3):** From Eq. (15), the relative uncertainty in Young's modulus satisfies:

$$\left(\frac{U_E}{E}\right)^2 \approx 9\left(\frac{U_L}{L}\right)^2 + \left(\frac{U_b}{b}\right)^2 + 9\left(\frac{U_h}{h}\right)^2 + \left(\frac{U_S}{S}\right)^2. \tag{22}$$

Because $E \propto 1/h^3$, fractional uncertainties in thickness are magnified by a factor of 3 in the modulus ($3U_h/h$). For Mild Steel, where vernier calliper readings across the rolled sheet showed thickness variations of $\Delta h = 0.14$ mm on an average $h = 2.41$ mm (5.8% relative variance), the propagated uncertainty in E expanded to $\pm$69.10 GPa (17.5%). Conversely, for Toughened Glass with tightly controlled float thickness ($h = 3.25 \pm 0.03$ mm, 0.9% variance), the relative uncertainty in E was only 3.4% ($\pm$7.33 GPa).

3. **Glass Elastic Modulus Discrepancy: Physical Origin of Systematic Error:** A critical physical finding is that thermal tempering does NOT alter the Young's modulus of glass. Both annealed float glass (NG) and thermally tempered glass (TG) share the identical amorphous silica network ($\mathrm{SiO_2 - Na_2O - CaO}$), whose bond-stretching stiffness is fundamentally invariant under thermal quenching. Both materials possess an identical theoretical elastic modulus of $E \approx$ 68–72 GPa (nominally 70 GPa).

However, the experimental results yield $E = 48.73 \pm 10.05$ GPa for NG and $E = 109.30 \pm 7.33$ GPa for TG. Their 95% confidence intervals overlap neither each other nor the true 70 GPa benchmark. This proves that the discrepancy is governed by systematic errors rather than random least-count uncertainty:

- **Cubic Thickness Sensitivity (h^3):** The measured thickness was $h = 3.87$ mm for NG versus $h = 3.25$ mm for TG. Their cubic ratio is $(3.87/3.25)^3 = 1.688$, creating an intrinsic 69% divergence in apparent modulus! Meanwhile, their experimental slopes in the linear regime are very close: $S_{\text{NG}} = 0.3872$ kN/mm versus $S_{\text{TG}} = 0.4453$ kN/mm (a ratio of only 1.15). A caliper measurement error of only 0.4–0.5 mm (arising from edge bevels, chamfers, or protective edge tape) completely shifts the calculated modulus: if the true thickness were $h = 3.45$ mm for NG and $h = 3.75$ mm for TG, both datasets would evaluate to exactly $E = 70.0$ GPa.
- **Toe Slack and Seating Compliance:** Raw NG data exhibited an initial ~ 0.14 mm region of zero-load travel and non-linear support seating before rigid line contact was established. Similarly, TG exhibited an initial concave compliance zone below 0.20 mm deflection. Zero-correcting and refitting in the truly linear regimes (0.25–0.55 mm for NG and 0.25–0.60 mm for TG) increased the NG modulus from 44.6 GPa to 48.7 GPa, but residual contact compliance remains.
- **Crosshead Compliance vs. Specimen Deflection:** The UTM LVDT measures crosshead displacement rather than direct optical beam deflection. Fixture compliance and localized cylindrical contact indentation under the loading pin add parasitic displacement ($\delta_{\text{measured}} = \delta_{\text{bending}} + \delta_{\text{machine}} + \delta_{\text{contact}}$), systematically lowering the apparent slope of stiff specimens.

4. **Early Test Termination and Absence of Plastic Yield in MS and PMMA:** The experimental tests for Mild Steel and Acrylic were terminated manually by the operators at small crosshead displacements ($\delta = 4.502$ mm for MS and $\delta = 8.858$ mm for Acrylic), far short of their ASTM validity limits ($\delta_{0.05} = 54.12$ mm and 61.16 mm).

   - **Mild Steel (MS):** The maximum outer fiber strain reached was only $\varepsilon_{f,\max} = 0.00417$ (0.42%). Because the standard 0.2% proof stress construction requires $\varepsilon \geq 0.002 + \sigma_y/E \approx 0.002 + 517/197580 \approx 0.0046$, the 0.2% offset line does not intersect the experimental curve within the recorded data (Figure 6.2). The reported value of 516.82 MPa is strictly the maximum flexural stress reached prior to test termination, NOT a yield strength.
   - **Acrylic (PMMA):** The maximum outer fiber strain reached was only $\varepsilon_{f,\max} = 0.00724$ (0.72%), corresponding to a maximum stress of 11.64 MPa. Glassy polymers undergo shear yielding at strains of 4–6% with yield strengths of 65–85 MPa. At 11.64 MPa, PMMA was deforming purely within its initial linear/viscoelastic regime and was far from yielding. Thus, 11.64 MPa cannot be compared against literature yield strength.

5. **Elevated Proof Strength in Copper (Cold-Worked Temper):** The experimental 0.2% proof yield strength of Copper was determined as 328.85 MPa, which significantly exceeds standard literature for annealed copper (70–220 MPa). This is explained by three distinct factors:

   - **Cold-Worked Metallurgical State:** The test strip was supplied in a cold-rolled temper (half-hard or hard copper, H02/H04) rather than dead-soft annealed condition. Cold rolling generates a high dislocation density ($\rho \sim 10^{15}$ m^{-2}) and dense dislocation cell walls, which elevates the flow stress via the Taylor relationship ($\sigma \propto \sqrt{\rho}$) to > 300 MPa.
   - **Flexural Strain-Gradient Constraint:** Unlike uniaxial tension where the entire gauge volume yields uniformly, flexural loading produces a steep linear stress gradient. The inner elastic core constrains the yielding of outer fibers, delaying the formation of a plastic hinge and shifting the apparent offset yield point upward.

- **Strain-Rate Sensitivity:** Group 1 conducted the test at an elevated crosshead speed ($v = 0.50$ mm/s vs. standard 0.01 mm/s), providing a modest dynamic increase in flow stress.

6. **Crosshead Compliance in PMMA and Copper Moduli:** The calculated Young's moduli for Acrylic (1.59 ± 1.11 GPa vs. lit. 2.4–3.4 GPa) and Copper (87.42 ± 18.17 GPa vs. lit. 110–128 GPa) fall below their bulk ultrasonic benchmarks. Because displacement was measured from the actuator crosshead LVDT, localized contact indentation of the loading nose into the compliant PMMA surface and thin copper strip, combined with load frame compliance, adds non-bending displacement that lowers the apparent tangent slope S.

7. **Weibull Statistics: Why Glass Is Stronger in Bending than Tension:** Uniaxial tensile tests on brittle glasses typically report tensile fracture strengths of only 20–35 MPa, whereas three-point bending yields 48.51 MPa for Normal Glass and 128.27 MPa for Toughened Glass. This discrepancy is explained by Weibull weakest-link statistics:

$$P_s(V) = \exp\left[-\int_V \left(\frac{\sigma}{\sigma_0}\right)^m \mathrm{d}V\right]. \tag{23}$$

In uniaxial tension, the entire specimen volume V experiences uniform maximum tensile stress σ. In three-point bending, the stress is non-uniform: it varies linearly from zero at the neutral axis to $\sigma_{\max}$ at the outer surface, and falls off linearly from midspan to the supports. The effective volume under peak tension is:

$$V_{\text{eff}} = \frac{V}{2(m+1)^2}. \tag{24}$$

For typical glass Weibull moduli ($m \approx 5$–8), the highly stressed volume in bending is less than 1% of the total specimen volume. Because the probability of finding a critical Griffith surface micro-crack within this tiny volume is dramatically lower, the apparent flexural fracture strength is significantly higher than the uniaxial tensile strength.

9 Conclusions

1. **Materials Elastic Spectrum and Glass Modulus Invariance:** The experimental elastic responses reflect the fundamental bonding spectrum of engineering materials, ranging from stiff metallic lattices (Mild Steel, $E = 197.58 \pm 69.10$ GPa) to compliant covalent polymers (Acrylic, $E = 1.59 \pm 1.11$ GPa). Crucially, both annealed normal glass and thermally tempered glass share an identical intrinsic Young's modulus ($E \approx 68$–72 GPa) governed by their identical silicate glass network. The apparent experimental divergence (48.73 GPa vs. 109.30 GPa) is a systematic metrological artifact driven by cubic thickness sensitivity ($E \propto h^{-3}$) and fixture seating slack.

2. **Validation of Beam Theory Identity:** For all six materials, the elastic modulus derived from the load–displacement slope ($E = \frac{L^3}{4bh^3}S$) was mathematically and numerically identical to the slope of the flexural stress–strain curve ($\mathrm{d}\sigma_f/\mathrm{d}\varepsilon_f$), fully confirming Euler–Bernoulli beam theory.

3. **Brittle Fracture and Tempering Strengthening:** Toughened glass exhibited a flexural fracture strength of 128.27 MPa, representing a 2.64× **strength enhancement** over annealed normal glass (48.51 MPa). This enhancement is quantitatively governed by the trapped compressive surface residual stress layer ($\Delta\sigma \approx 79.8$ MPa) produced by thermal quenching.

4. **Rate Sensitivity Contrast:** Copper displays low rate sensitivity ($m \approx 0.012$) due to athermal dislocation forest interactions, maintaining an athermal elastic modulus and stable flow stress. Conversely, Acrylic exhibits strong rate dependence ($m \approx 0.07$) governed by cooperative segmental viscoelastic relaxation and ester side-group rotation (β-transition).

5. **Critical Assessment of Literature Agreement:** Rather than uniform agreement, rigorous analysis reveals distinct experimental and metallurgical insights:

   - Only Mild Steel modulus (197.58 ± 69.10 GPa) falls squarely within the literature range (205–210 GPa).
   - Copper (87.42 GPa) and Acrylic (1.59 GPa) moduli are moderately underestimated relative to bulk literature (110–128 GPa and 2.4–3.4 GPa) due to crosshead LVDT compliance and local roller contact indentation.
   - Copper proof yield strength (328.85 MPa) exceeds annealed literature (70–220 MPa) because the test specimen was in a cold-worked rolled strip temper with flexural plastic constraint.
   - Mild Steel (516.82 MPa) and Acrylic (11.64 MPa) tests were manually stopped before reaching plastic yield, so these values represent maximum stresses reached rather than material yield strengths.
   - Normal and Toughened Glass moduli diverge from the true 70 GPa baseline due to extreme h^3 metrological sensitivity and fixture seating slack.

References

[1] ASTM D790-17, *Standard Test Methods for Flexural Properties of Unreinforced and Reinforced Plastics and Electrical Insulating Materials*, ASTM International, West Conshohocken, PA, 2017.

[2] ASTM C158-02, *Standard Test Methods for Strength of Glass by Flexure (Determination of Modulus of Rupture)*, ASTM International, 2017.

[3] W. D. Callister and D. G. Rethwisch, *Materials Science and Engineering: An Introduction*, 10th ed., John Wiley & Sons, Hoboken, NJ, 2018.

[4] G. E. Dieter, *Mechanical Metallurgy*, 3rd ed., McGraw-Hill, New York, 1986.

[5] R. W. Hertzberg, R. P. Vinci, and J. L. Hertzberg, *Deformation and Fracture Mechanics of Engineering Materials*, 5th ed., John Wiley & Sons, 2012.

[6] M. A. Meyers and K. K. Chawla, *Mechanical Behavior of Materials*, 2nd ed., Cambridge University Press, Cambridge, 2009.

[7] M. F. Ashby, *Materials Selection in Mechanical Design*, 4th ed., Butterworth-Heinemann, Oxford, 2011.

A Appendix

A.1 Raw Data Files and Python Post-Processing

- **Experimental Data Source:** `GP1 FRI 3PT BEND.xlsx` (Sheet: `Sheet1`, Columns 1–17 covering MS, Al, Cu, Acrylic, NG, TG).
- **Laboratory Courseware Reference:** Lab Handout `3-POINT BEND TEST - 2026.pdf` and course lecture slides (Prof. S. Karthikeyan, IISc Bangalore).

- **Code Pipeline:** Python 3 script `process_and_plot.py` utilizing `openpyxl`, `numpy`, `scipy`, and `matplotlib`.

A.2 Sample Calculations: Mild Steel (MS)

1. **Area Moment of Inertia ($\bar{I}$):**

$$\bar{I} = \frac{1}{12} b h^3 = \frac{1}{12} \times 29.92 \text{ mm} \times (2.41 \text{ mm})^3 = \frac{29.92 \times 13.9975}{12} = 34.899 \text{ mm}^4.$$

2. **Validity Displacement Limit ($\delta_{0.05}$):**

$$\delta_{0.05} = \frac{0.05 L^2}{6h} = \frac{0.05 \times (125.0 \text{ mm})^2}{6 \times 2.41 \text{ mm}} = \frac{781.25}{14.46} = 54.03 \text{ mm}.$$

3. **Elastic Tangent Slope (S):** Linear regression over $\delta \in [0.20, 1.50]$ mm ($N = 651$ points):

$$S = \frac{\mathrm{d}P}{\mathrm{d}\delta} = 0.1695 \text{ kN/mm} = 169.5 \text{ N/mm}, \qquad R^2 = 0.99990, \qquad \text{RMS} = 0.65 \text{ N}.$$

4. **Young's Modulus (E):**

$$\begin{aligned} E &= \frac{L^3}{4bh^3} S = \frac{(125.0 \text{ mm})^3}{4 \times 29.92 \text{ mm} \times (2.41 \text{ mm})^3} \times 169.5 \text{ N/mm} \\ &= \frac{1,953,125}{1675.16} \times 169.5 = 197,580 \text{ MPa} = 197.58 \text{ GPa}. \end{aligned}$$

5. **Maximum Flexural Stress at Test Stop ($P = 0.479$ kN $= 479.0$ N, No Yield Reached):**

$$\sigma_{f,\max} = \frac{3PL}{2bh^2} = \frac{3 \times 479.0 \text{ N} \times 125.0 \text{ mm}}{2 \times 29.92 \text{ mm} \times (2.41 \text{ mm})^2} = \frac{179,625}{347.56} = 516.82 \text{ MPa}.$$

   Note on Yield Status: Because the test was terminated at $\delta = 4.502$ mm ($\varepsilon_{f,\max} = 0.00417$), the 0.2% offset construction line ($\sigma = E(\varepsilon - 0.002)$) did not intersect the curve. Thus, 516.82 MPa is the maximum stress reached prior to test termination, not a yield strength.

6. **Maximum Flexural Strain at Test Stop ($\delta = 4.502$ mm):**

$$\varepsilon_{f,\max} = \frac{6\delta h}{L^2} = \frac{6 \times 4.502 \text{ mm} \times 2.41 \text{ mm}}{(125.0 \text{ mm})^2} = \frac{65.10}{15625} = 0.004166 \quad (0.417\%).$$

A.3 Sample Calculations: Toughened Glass (TG)

1. **Centroidal Area Moment of Inertia ($\bar{I}$):**

$$\bar{I} = \frac{1}{12} b h^3 = \frac{1}{12} \times 29.67 \text{ mm} \times (3.25 \text{ mm})^3 = \frac{29.67 \times 34.328}{12} = 84.877 \text{ mm}^4.$$

2. **Elastic Tangent Slope (S):** Linear regression over truly linear region $\delta \in [0.25, 0.60]$ mm ($N = 175$ points):

$$S = \frac{\mathrm{d}P}{\mathrm{d}\delta} = 0.4453 \text{ kN/mm} = 445.3 \text{ N/mm}, \qquad R^2 = 0.99989, \qquad \text{RMS} = 0.47 \text{ N}.$$

3. **Young's Modulus (E):**

$$E = \frac{L^3}{4bh^3} S = \frac{(100.0\ \text{mm})^3}{4 \times 29.67\ \text{mm} \times (3.25\ \text{mm})^3} \times 445.3\ \text{N/mm}$$
$$= \frac{1,000,000}{4074.11} \times 445.3 = 109,300\ \text{MPa} = 109.30\ \text{GPa}.$$

4. **Flexural Fracture Strength (Modulus of Rupture) at $P_{\text{fracture}} = 268.0$ N:**

$$\sigma_{f,\text{fracture}} = \frac{3P_{\text{fracture}}L}{2bh^2} = \frac{3 \times 268.0\ \text{N} \times 100.0\ \text{mm}}{2 \times 29.67\ \text{mm} \times (3.25\ \text{mm})^2} = \frac{80,400}{626.78} = 128.27\ \text{MPa}.$$

5. **Fracture Strain at $\delta = 0.652$ mm:**

$$\varepsilon_{f,\text{fracture}} = \frac{6\delta h}{L^2} = \frac{6 \times 0.652\ \text{mm} \times 3.25\ \text{mm}}{(100.0\ \text{mm})^2} = \frac{12.714}{10000} = 0.001271 \quad (0.127\%).$$

A.4 Uncertainty Propagation Derivations

From Eq. (9):

$$E = \frac{L^3}{4bh^3} S.$$

Taking the natural logarithm:

$$\ln E = 3\ln L - \ln 4 - \ln b - 3\ln h + \ln S.$$

The total differential is:

$$\frac{\mathrm{d}E}{E} = 3\frac{\mathrm{d}L}{L} - \frac{\mathrm{d}b}{b} - 3\frac{\mathrm{d}h}{h} + \frac{\mathrm{d}S}{S}.$$

Assuming independent, random, Gaussian uncertainties, the combined standard uncertainty $u_c(E)$ is given by the root-sum-square:

$$\frac{u_c(E)}{E} = \sqrt{\left(3\frac{u(L)}{L}\right)^2 + \left(\frac{u(b)}{b}\right)^2 + \left(3\frac{u(h)}{h}\right)^2 + \left(\frac{u(S)}{S}\right)^2}.$$

The expanded uncertainty reported at the 95% confidence level ($k = 2$) is:

$$U_E = 2 \cdot u_c(E) = 2E\sqrt{9\left(\frac{u(L)}{L}\right)^2 + \left(\frac{u(b)}{b}\right)^2 + 9\left(\frac{u(h)}{h}\right)^2 + \left(\frac{u(S)}{S}\right)^2}.$$